Inner Product Spaces

For vectors (in vector space over a field)

(I) addition, scaling ✓

(II) Angle

↳ angle measurement through inner product

↓ +ve definiteness

length (norm)

(possible when field is ordered, $\mathbb{R}$ or $\mathbb{C}$).

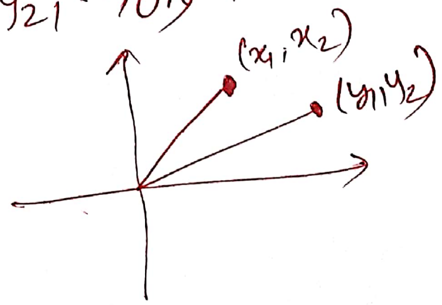
Inner products

The length of a vector $\bar{v} \in \mathbb{R}^n$ is called the norm of $\bar{v}$, denoted as $\|\bar{v}\|$,

$$\|\bar{v}\| := \sqrt{\sum_{i=1}^{n} x_i^2}, \text{ where } \bar{v} = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n.$$

Definition Dot product: For $\bar{x}, \bar{y} \in \mathbb{R}^n$, the dot product of $\bar{x}$ & $\bar{y}$ is

$$\bar{x} \cdot \bar{y} = \sum_{i=1}^{n} x_i y_i, \text{ where } \bar{x} = (x_1, x_2, \dots, x_n),\ \bar{y} = (y_1, y_2, \dots, y_n).$$



Note: $\bar{x} \cdot \bar{x} = \|\bar{x}\|^2 \quad \forall \bar{x} \in \mathbb{R}^n$.

Properties of dot product on $\mathbb{R}^n$:

(I) $\bar{x} \cdot \bar{x} \geq 0 \quad \forall x \in \mathbb{R}^n$.

(II) $\bar{x} \cdot \bar{x} = 0$ iff $\bar{x} = \bar{0}$.

(III) for $\bar{y} \in \mathbb{R}^n$ fixed, the map from $\mathbb{R}^n \to \mathbb{R}$ that sends $\bar{x} \in \mathbb{R}^n$ to $\bar{x} \cdot \bar{y}$ is linear.

(IV) $\bar{x} \cdot \bar{y} = \bar{y} \cdot \bar{x} \quad \forall \bar{x}, \bar{y} \in \mathbb{R}^n$.

For a complex no. $\lambda = a + ib$, $a, b \in \mathbb{R}$, then

(i) The absolute value of λ, $|\lambda| = \sqrt{a^2+b^2}$;

(ii) the complex conjugate of λ, $\bar{\lambda} = a - ib$;

(iii) $|\lambda|^2 = \lambda\bar{\lambda}$.

For $\bar{z} = (z_1, z_2, \ldots, z_n) \in \mathbb{C}^n$, the norm of $\bar{z}$

$$\|\bar{z}\| = \sqrt{\sum_{i=1}^{n} |z_i|^2} = \sqrt{\sum_{i=1}^{n} z_i \cdot \bar{z}_i}.$$

An inner product on V (over F, where F is $\mathbb{R}$ or $\mathbb{C}$) is a f^n that takes each ordered pair $(\bar{u}, \bar{v})$ of V, for $\bar{u} \in V, \bar{v} \in V$, to a scalar $\langle \bar{u}, \bar{v} \rangle \in F$ and has the following properties:

(i) +vity $\langle \bar{v}, \bar{v} \rangle \geq 0 \ \forall \ \bar{v} \in V$;

(ii) definiteness $\langle \bar{v}, \bar{v} \rangle = 0$ iff $\bar{v} = \bar{0}$.

(iii) additivity in first argument

$\langle \bar{u} + \bar{v}, \bar{w} \rangle = \langle \bar{u}, \bar{w} \rangle + \langle \bar{v}, \bar{w} \rangle \ \forall \ \bar{u}, \bar{v}, \bar{w} \in V$.

(iv) homogenity in first argument

$\langle \lambda\bar{u}, \bar{v} \rangle = \lambda \langle \bar{u}, \bar{v} \rangle \ \forall \ \lambda \in F \ \& \ \forall \bar{u}, \bar{v} \in V$.

(v) conjugate symmetry

$\langle \bar{u}, \bar{v} \rangle = \overline{\langle \bar{v}, \bar{u} \rangle} \ \forall \ \bar{u}, \bar{v} \in V$.

Examples:

(a) Euclidean inner product on $\mathbb{F}^n$ is

$$\langle (w_1, \dots, w_n), (z_1, \dots, z_n) \rangle = w_1 \bar{z}_1 + \dots + w_n \bar{z}_n .$$

(b) If $c_1, \dots, c_n \geq 0$ (+ve nos.), an inner product can be defined on $\mathbb{F}^n$ by

$$\langle (w_1, w_2, \dots, w_n), (z_1, z_2, \dots, z_n) \rangle = c_1 w_1 \bar{z}_1 + \dots + c_n w_n \bar{z}_n .$$

(c) For vector space of continuous real-valued f^ns on interval $[-1, 1]$, an inner product can be defined as

$$\langle f, g \rangle = \int_{-1}^{1} f(x)\, g(x)\, dx .$$

(d) An inner product on $\mathcal{P}(\mathbb{R})$ can be defined as $\langle p, q \rangle = \int_0^{\infty} p(x)\, q(x)\, e^{-x}\, dx .$

An inner product space is a vector space V along w/ an inner product on V.

→ $\mathbb{F}^n$ is an inner product space, by default we assume inner product to be Eulidean.

→ By vector space V, we are going to mean it is an inner product space over F by default if F is $\mathbb{R}$ or $\mathbb{C}$, which is the case for all topics in this semester from now on.

Basic properties of an inner product

ⓐ For each fixed $\bar{u} \in V$, the f! that takes $\bar{v}$ to $\langle \bar{v}, \bar{u} \rangle$ is a linear map from V to F.

ⓑ $\langle \bar{0}, \bar{u} \rangle = 0 \quad \forall \, \bar{u} \in V$.

ⓒ $\langle \bar{u}, 0 \rangle = 0 \quad \forall \, \bar{u} \in V$.

ⓓ $\langle \bar{u}, \bar{v} + \bar{w} \rangle = \langle \bar{u}, \bar{v} \rangle + \langle \bar{u}, \bar{w} \rangle$
$\forall \, \bar{u}, \bar{v}, \bar{w} \in V$.

ⓔ $\langle \bar{u}, \lambda \bar{v} \rangle = \bar{\lambda} \langle \bar{u}, \bar{v} \rangle \quad \forall \, \lambda \in F, \forall \, \bar{u}, \bar{v} \in V$.

Proof:

For $\bar{v} \in V$, the norm of $\bar{v}$,

$$\|\bar{v}\| = \sqrt{\langle v, v \rangle}.$$

Basic properties of the norm:

Suppose $\bar{v} \in V$,

ⓐ $\|\bar{v}\| = 0$ iff $\bar{v} = \bar{0}$.

ⓑ $\|\lambda \bar{v}\| = |\lambda| \, \|\bar{v}\| \quad \forall \lambda \in F$.

Two vectors $\bar{v}, \bar{u} \in V$ are called orthogonal if $\langle \bar{u}, \bar{v} \rangle = \bar{0}$, ~~there $\bar{v}, \bar{u} \neq \bar{0}$.~~

Orthogonality & $\bar{0}$

ⓐ $\bar{0}$ is orthogonal to every $\bar{v} \in V$.

ⓑ $\bar{0}$ is the only vector in V that is orthogonal to itself.

Prove that for $\bar{u}, \bar{v} \in \mathbb{R}^2$, $\langle \bar{u}, \bar{v} \rangle = \|\bar{u}\| \|\bar{v}\| \cos\theta$, where θ is angle bet^n $\bar{u}, \bar{v}$.

Let $\bar{u}, \bar{v}$ be orthogonal vectors in V. Then

$$\|\bar{u}+\bar{v}\|^2 = \|\bar{u}\|^2 + \|\bar{v}\|^2.$$

Proof: $\|\bar{u}+\bar{v}\|^2 = \langle \bar{u}+\bar{v}, \bar{u}+\bar{v} \rangle = \|\bar{u}\|^2 + \|\bar{v}\|^2 + \cancel{2\mathrm{Re}} \langle \bar{u}, \bar{v} \rangle + \langle \bar{v}, \bar{u} \rangle$

$= \|\bar{u}\|^2 + \|\bar{v}\|^2.$

(What happens when $\langle \bar{u}, \bar{v} \rangle + \langle \bar{v}, \bar{u} \rangle = \bar{0}$?,

when is $\langle \bar{u}, \bar{v} \rangle + \langle \bar{v}, \bar{u} \rangle = 2\,\mathrm{Re}\langle \bar{u}, \bar{v} \rangle = 0$?

In real inner product space $\langle \bar{u}, \bar{v} \rangle = 0$

$\Rightarrow \bar{u} \perp \bar{v}$).

To write $\bar{u} = c\bar{v} + \bar{v}^{\perp}$, where $\bar{v}^{\perp} \perp \bar{v}$, for $c \in F$, note that $\bar{v}^{\perp} = \bar{u} - c\bar{v}$.

Then, $\bar{u} = c\bar{v} + \bar{u} - c\bar{v}$. To find $c \in F$,

$\langle \bar{u} - c\bar{v}, \bar{v} \rangle = 0 = \langle \bar{u}, \bar{v} \rangle - c\|\bar{v}\|^2$

or, $c = \dfrac{\langle \bar{u}, \bar{v} \rangle}{\|\bar{v}\|^2}$.

That is, $\bar{u} = \dfrac{\langle \bar{u}, \bar{v} \rangle}{\|\bar{v}\|^2}\bar{v} + \left(u - \dfrac{\langle \bar{u}, \bar{v} \rangle}{\|\bar{v}\|^2}\bar{v} \right)$.

Thm. Let $\bar{u}, \bar{v} \in V$, where $\bar{v} \neq 0$. For $c = \dfrac{\langle \bar{u}, \bar{v} \rangle}{\|\bar{v}\|^2}$

and $\bar{w} = \bar{u} - \dfrac{\langle \bar{u}, \bar{v} \rangle}{\|\bar{v}\|^2}\bar{v}$. Then,

$\langle \bar{w}, \bar{v} \rangle = 0$ & $\bar{u} = c\bar{v} + \bar{w}$.

Thm. (Cauchy-Schwarz inequality)

Let $\bar{u}, \bar{v} \in V$. Then, $|\langle \bar{u}, \bar{v} \rangle| \leq \|\bar{u}\| \cdot \|\bar{v}\|$.

The inequality is saturated ("=" holds) iff one of $\bar{u}, \bar{v}$ is scalar multiple of the other.

Proof: Trivially holds for $\bar{v} = \bar{0}$.

For $\bar{v} \neq \bar{0}$, $\bar{u} = \dfrac{\langle \bar{u}, \bar{v} \rangle}{\|\bar{v}\|^2}\bar{v} + \bar{v}^{\perp}$, where $\bar{v}^{\perp} \perp \bar{v}$.

$$\|\bar{u}\|^2 = \left\| \frac{\langle \bar{u}, \bar{v} \rangle}{\|\bar{v}\|^2}\bar{v} \right\|^2 + \|\bar{v}^{\perp}\|^2 = \frac{|\langle \bar{u}, \bar{v} \rangle|^2}{\|\bar{v}\|^2} + \|\bar{v}^{\perp}\|^2$$

$\geq \dfrac{|\langle \bar{u}, \bar{v} \rangle|^2}{\|\bar{v}\|^2}$, and Inequality saturates iff $\|\bar{v}^{\perp}\| = 0 \Leftrightarrow \bar{v}^{\perp} = \bar{0}$.

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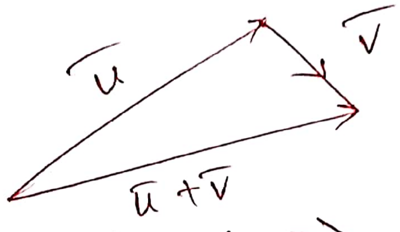
Thm ($\triangle$ gle inequality)

Let $\bar{u}, \bar{v} \in V$. Then, $\|\bar{u}+\bar{v}\| \leq \|\bar{u}\| + \|\bar{v}\|$.
Inequality saturates iff one of $\bar{u}, \bar{v}$ is a nonnegative multiple of the other.

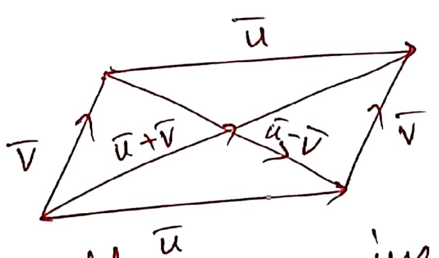


Proof:

$$\|\bar{u}+\bar{v}\|^2 = \langle \bar{u}+\bar{v}, \bar{u}+\bar{v} \rangle$$
$$= \langle \bar{u},\bar{u}\rangle + \langle \bar{v},\bar{v}\rangle + \langle \bar{u},\bar{v}\rangle + \langle \bar{v},\bar{u}\rangle$$
$$= \|\bar{u}\|^2 + \|\bar{v}\|^2 + 2\,\mathrm{Re}\,\langle \bar{u},\bar{v}\rangle$$
$$\leq \|\bar{u}\|^2 + \|\bar{v}\|^2 + 2\,|\langle \bar{u},\bar{v}\rangle|$$
$$\leq \|\bar{u}\|^2 + \|\bar{v}\|^2 + 2\,\|\bar{u}\|\|\bar{v}\| = (\|\bar{u}\| + \|\bar{v}\|)^2.$$

using Cauchy-Schwarz ineq.

Inequality saturates iff $\langle \bar{u},\bar{v}\rangle = \|\bar{u}\|\|\bar{v}\|$.



Thm. (Parallelogram inequality)

Let $\bar{u}, \bar{v} \in V$. Then, $\|\bar{u}+\bar{v}\|^2 + \|\bar{u}-\bar{v}\|^2 = 2(\|\bar{u}\|^2 + \|\bar{v}\|^2)$.

Proof: